

Video-Based Road Damage Detection

Using Image Enhancement and Segmentation

Group 03 | 2nd Year Undergraduate | Department of Information Technology

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Problem Statement

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Road Damage is Widespread

Sri Lankan roads face potholes, cracks, patch failures, and edge cracking due to heavy traffic and adverse weather.

Manual Inspection is Inadequate

Current assessments are expensive, labour-intensive, time-consuming, and subjective — inconsistent results across inspectors.

Scale Challenge

Monitoring large national road networks manually is impractical and unsustainable.

Why It Matters

- **Vehicle Damage**

Increased operating costs

- **Safety Risk**

Accidents due to poor roads

- **Economic Loss**

Maintenance budget inefficiency

- **Sri Lanka Focus**

Mihintale, Anuradhapura, residential roads

Project Objectives

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01 Develop a Video-Based System

Design and implement an image processing pipeline that detects road surface damages from mobile video footage.

02 Apply Frame Enhancement

Use grayscale conversion, noise reduction, CLAHE contrast enhancement, illumination correction, and edge sharpening to improve frame quality.

03 Perform Reliable Segmentation

Apply Black-Hat morphological transform, binary thresholding, morphological closing, and connected component analysis to isolate damage regions.

04 Evaluate System Performance

Measure accuracy, precision, recall, F1-score, and IoU against ground truth; validate under varying environmental conditions.

Video Acquisition Method

Recording Device

Apple iPhone 11

Resolution: 1920×1080 (Full HD)

Frame Rate: 30 FPS

Duration: 15–30 sec per clip

Recording Sites

Mihintale • Kanadarawa • Anuradhapura New Town
Residential roads (tar & concrete)
Urban, semi-urban, and rural environments

Ethical Compliance: No identifiable individuals or licence plates captured. Recordings in public areas only.

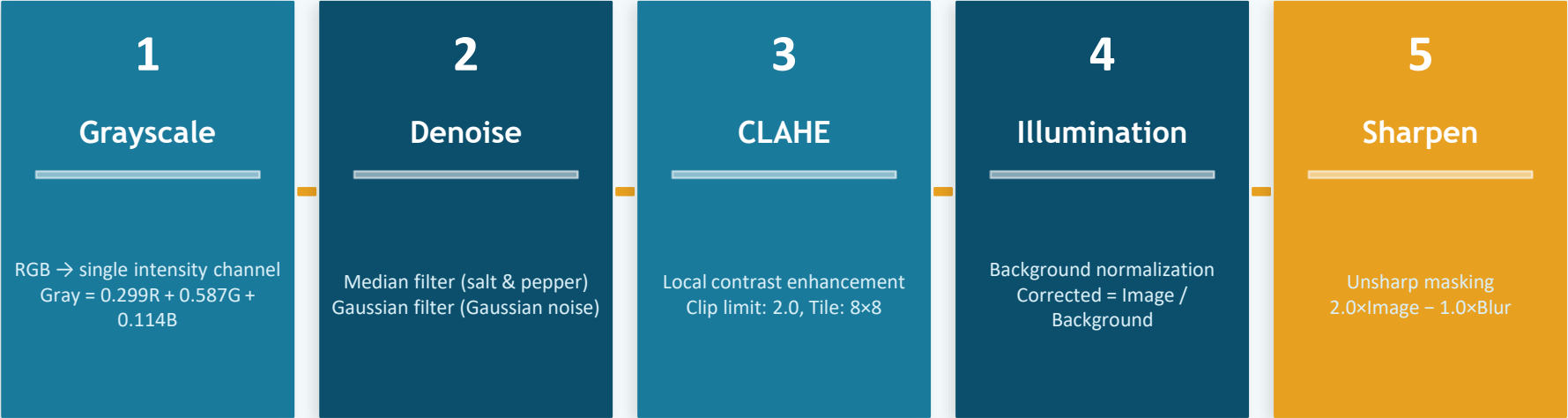
Damage Types Captured

- Potholes
- Surface Cracks
- Patch Failures
- Edge Cracking
- Road Marking Removal

Frame Extraction Process

Every 10th frame extracted (uniform sampling – reduces redundancy)
Saved as JPEG • Original resolution preserved
Annotated with Labelling (bounding box ground truth)

Frame Enhancement Pipeline

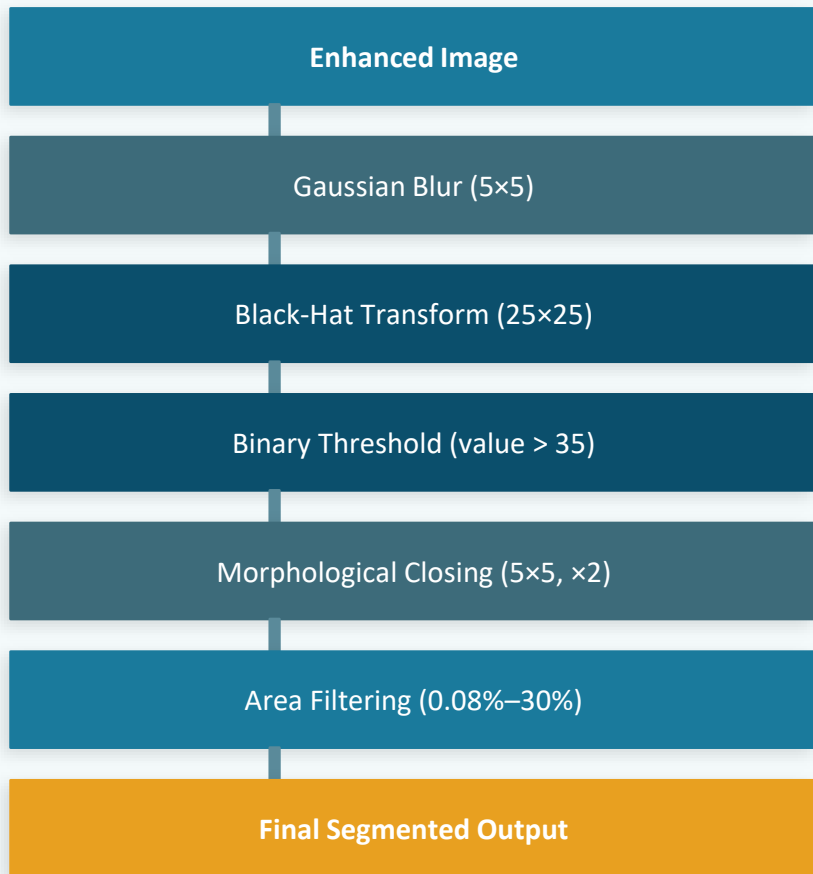


Quantitative Improvement (Sample Frame)

Metric	Original Frame	Enhanced Frame	Improvement
Intensity Range	120	198	+65%
Laplacian Variance (Sharpness)	85.3	162.7	+91%
Mean Intensity	92.4	110.6	+20%

Segmentation Pipeline

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Black-Hat Transform

$$T_{blackhat} = Closing(I) - I$$

Isolates dark regions SMALLER than a 25x25 kernel → targets cracks/potholes, ignores large shadows.

Morphological Closing

$$Closing(A,B) = (A \oplus B) \ominus B$$

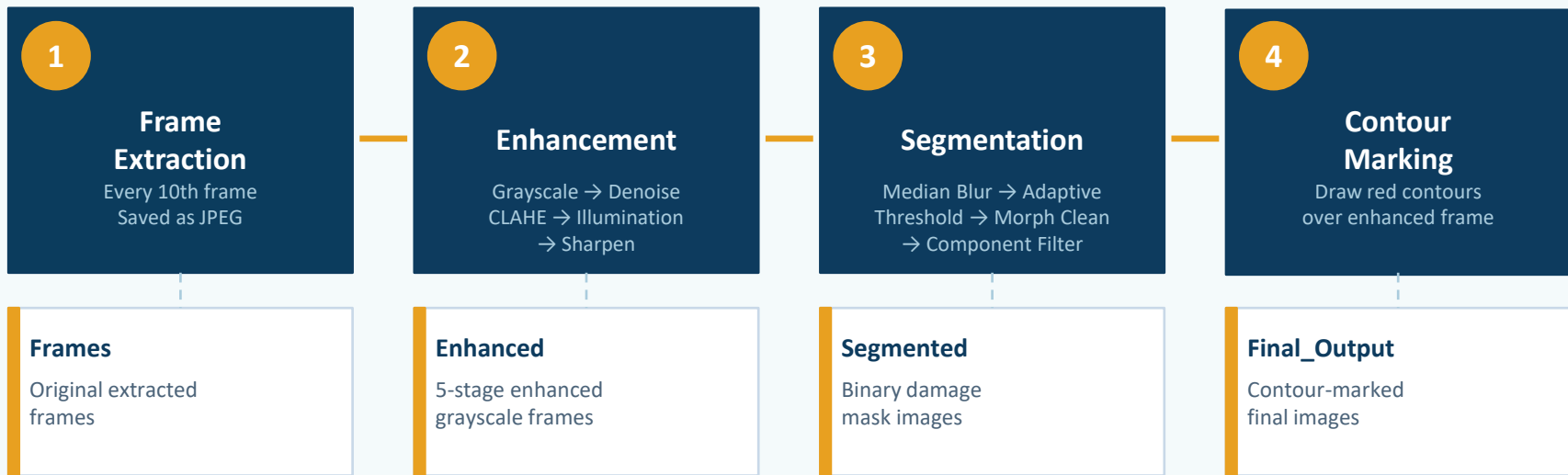
Bridges fragmented crack segments, ensuring structural continuity for accurate severity assessment.

Dynamic Area Filtering

Accepts contours between 0.08% and 30% of frame area.
Eliminates gravel noise (too small) and false shadow regions (too large).

Integrated Processing Pipeline

4-Stage End-to-End Video Processing



Key Pipeline Loop (integrated_code.py)

```
while True:
    frame = cap.read()          # extract
    enhanced = enhance_frame(frame) # enhance
    mask = get_segmentation_mask(...) # segment
    marked = draw_damage_contours(...) # mark & save
```

Pipeline Statistics

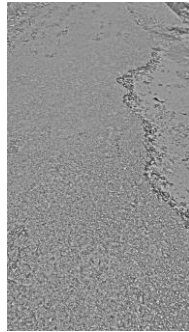
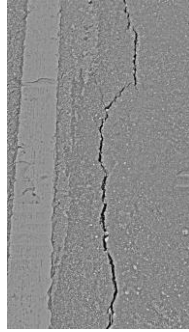
Frame Interval:	Every 10th frame
Output Folders:	4 (Frames / Enhanced / Segmented / Final)
Format:	JPEG · Original resolution preserved
Processing:	Sequential per-frame pipeline

Visual Comparison

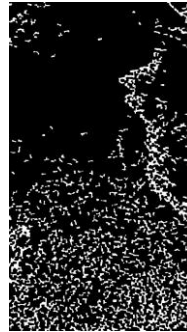
Original Image



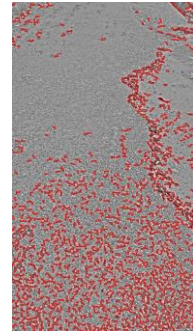
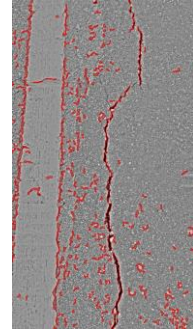
Enhanced Image



Segmented Image



Final Output

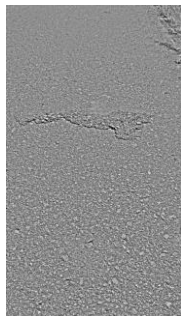


Cont...

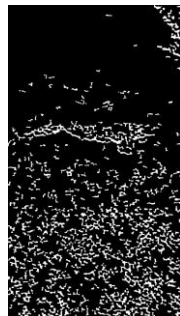
Original Image



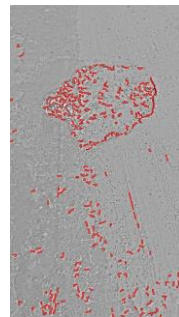
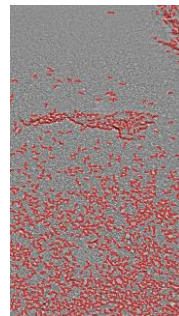
Enhanced Image



Segmented Image



Final Output



Experimental Setup

Dataset Summary

Video	Damage Type	Frames Extracted	Frames Tested	Conditions
V1	Crack	100	30	Mixed lighting
V2	Pothole	120	35	Bright + shadow
V3	Patch Failure	90	25	Uneven illumination
V4	Edge Cracking	70	30	Shadow + low contrast
Total	4 types	380	120	Real-world variation

Enhancement Parameters

Gaussian Kernel	5×5
Median Window	5×5
CLAHE Clip Limit	2.0
CLAHE Tile Size	8×8
Illumination Kernel	71×71
Sharpen (α / β)	2.0 / -1.0

Segmentation Parameters

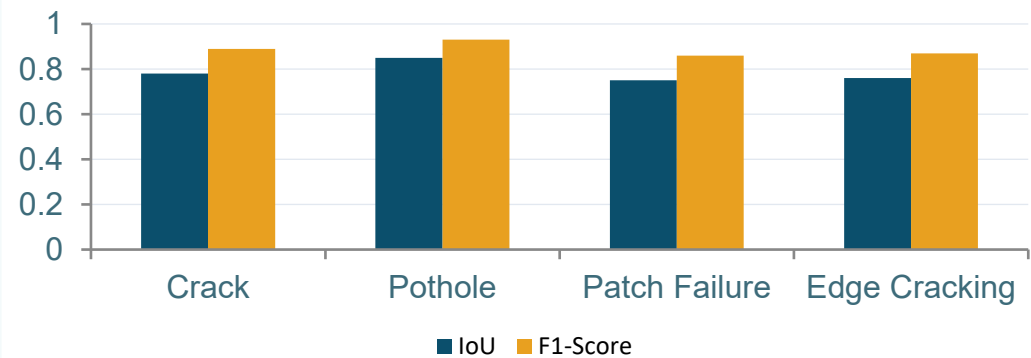
Enhancement Blur	5×5 Gaussian
Black-Hat Kernel	25×25 rectangle
Binary Threshold	Pixel value > 35
Morphological Close	5×5, 2 iterations
Min Area Filter	0.08% of frame
Max Area Filter	30% of frame

Results & Evaluation

Segmentation Performance by Damage Type (120 test frames)

Damage Type	Accuracy	Precision	Recall	F1-Score	IoU
Crack	0.92	0.90	0.84	0.89	0.78
Pothole	0.95	0.94	0.90	0.93	0.85
Patch Failure	0.90	0.88	0.81	0.86	0.75
Edge Cracking	0.91	0.89	0.80	0.87	0.76

IoU & F1-Score by Damage Type



Key Findings

Potholes achieved highest IoU (0.85) – strong boundary detection

Black-Hat transform successfully eliminated shadow false positives

Dynamic area filter removed gravel noise while preserving thin cracks

Average accuracy >90% across all damage categories

Limitations

Extreme Shadows

Very dark shadows cannot be fully removed by normalization — residual dark patches may be mistaken for cracks during segmentation.

Wet Road Conditions

Surface reflections from rain alter pixel intensity profiles, potentially causing false detections or missed damage regions.

Motion Blur

Excessive camera shake or fast movement blurs crack edges into road texture, reducing recall in blur-heavy frames.

Compression Artifacts

Block noise from MP4 video encoding can persist after filtering, introducing unwanted texture patterns.

Non-Structural Anomalies

Dark tire marks, oil spills, or wet patches mimic pothole intensity profiles, occasionally causing false positives.

Patch Boundary Over-segmentation

The algorithm sometimes segments patch boundaries as multiple individual cracks rather than a single unified region.

Conclusion & Future Work

Conclusions

- A complete video-based road damage detection system was successfully designed and implemented using mobile footages from Sri Lankan roads.
- The 5-stage enhancement pipeline consistently improved image quality — intensity range +65%, sharpness +91%.
- The Black-Hat segmentation approach eliminated shadow-based false positives that global thresholding (Otsu) could not handle.
- Average segmentation accuracy exceeded 90% across four damage categories with potholes achieving IoU of 0.85.
- The system provides a scalable, low-cost alternative to manual road inspection.

Future Work

Deep Learning Integration

Use CNN/YOLO-based models for real-time classification

Temporal Consistency

Frame-to-frame tracking to filter transient noise

Adaptive Parameters

Dynamic area thresholds based on camera perspective

Shape-Based Filtering

Eccentricity/circularity to distinguish cracks from oil spots

Advanced Enhancement

Deep learning-based denoising for extreme conditions

THANK YOU !